

BRATTMON, Sweden

WMO: 024130

Lat: 60.7569N

Lon: 12.7723E

Elev: 199

StdP: 98.96

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -24.2	(c) -21.5	(d) -26.3	(e) 0.3	(f) -24.1	(g) -23.3	(h) 0.5	(i) -21.4	(j) 4.8	(k) 1.6	(l) 4.3	(m) 1.9	(n) 0.8	(o) 320	(p) 0.590

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.1	(c) 27.1	(d) 17.5	(e) 25.1	(f) 16.7	(g) 23.0	(h) 15.6	(i) 18.7	(j) 24.7	(k) 17.6	(l) 23.0	(m) 16.6	(n) 21.4	(o) 2.2	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.7	12.2	20.1	15.6	11.4	19.1	14.7	10.7	18.2	53.6	24.9	50.2	23.0	47.4	21.5	21.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4.4	3.9	3.3	-27.0	29.3	2.7	1.7	-28.9	30.6	-30.4	31.6	-31.9	32.5	-33.9	33.8
			-27.2	20.0	2.5	0.9	-29.0	20.7	-30.4	21.2	-31.8	21.7	-33.7	22.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.0	-7.8	-6.0	-1.1	4.2	8.7	13.1	15.7	14.4	9.8	4.2	-0.7	-6.7	(6)
(7)		DBStd	9.28	6.86	5.53	4.65	3.09	3.62	2.99	2.69	2.76	3.07	4.10	4.95	7.35	(7)
(8)		HDD10.0	2659	551	447	343	176	68	6	0	2	41	184	322	519	(8)
(9)		HDD18.3	5236	809	681	602	425	298	161	91	124	257	439	572	777	(9)
(10)		CDD10.0	478	0	0	0	1	28	98	176	138	33	3	0	0	(10)
(11)		CDD18.3	13	0	0	0	0	0	2	8	2	0	0	0	0	(11)
(12)		CDH23.3	367	0	0	0	0	19	103	187	58	1	0	0	0	(12)
(13)		CDH26.7	63	0	0	0	0	2	20	37	4	0	0	0	0	(13)
(14)	Wind	WSAvg	1.4	1.2	1.2	1.5	1.6	1.7	1.6	1.5	1.3	1.3	1.3	1.3	1.1	(14)
(15)	Precipitation	PrecAvg	736	50	35	31	41	61	81	86	97	70	72	62	50	(15)
(16)		PrecMax	1011	119	61	66	102	185	150	187	146	128	158	172	93	(16)
(17)		PrecMin	578	11	7	5	9	4	8	30	35	24	15	21	5	(17)
(18)		PrecStd	90	25	14	14	25	38	34	32	33	26	32	35	24	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.3	7.0	14.3	19.5	25.9	29.0	29.8	27.3	21.8	15.3	11.1	9.1	(19)
(20)			MCWB	4.5	4.7	7.7	10.9	15.7	17.9	18.7	17.6	14.4	11.4	9.4	7.0	(20)
(21)		2%	DB	4.2	4.7	10.5	17.0	22.6	26.4	27.7	25.0	19.2	13.7	9.6	6.8	(21)
(22)			MCWB	2.7	2.7	5.3	9.9	13.6	16.5	18.3	17.0	13.7	10.8	7.7	5.5	(22)
(23)		5%	DB	2.5	2.8	7.7	14.5	20.1	23.7	25.7	23.0	17.3	12.0	7.8	4.6	(23)
(24)			MCWB	1.5	1.3	3.8	8.1	12.2	14.9	17.3	16.1	12.6	9.9	6.4	3.4	(24)
(25)		10%	DB	1.0	1.3	5.5	12.0	16.9	21.3	23.3	21.0	15.7	10.6	6.1	2.6	(25)
(26)			MCWB	0.3	0.4	2.5	6.7	10.1	13.9	16.3	15.6	11.9	8.9	5.1	1.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.7	4.8	8.0	12.1	16.4	19.0	20.1	19.6	15.9	12.8	9.8	7.3	(27)
(28)			MCDWB	6.0	6.5	13.5	18.0	23.6	26.7	27.5	24.2	20.0	14.4	11.1	8.8	(28)
(29)		2%	WB	3.0	3.2	5.7	10.1	14.5	17.3	19.0	18.1	14.6	11.3	8.0	5.5	(29)
(30)			MCDWB	3.9	4.5	9.7	16.3	21.1	24.9	25.9	22.4	17.5	12.8	9.4	6.6	(30)
(31)		5%	WB	1.6	1.6	4.2	8.6	12.8	16.0	18.0	17.1	13.5	10.1	6.6	3.6	(31)
(32)			MCDWB	2.3	2.5	7.0	13.5	18.0	22.1	23.9	21.2	16.3	11.6	7.7	4.6	(32)
(33)		10%	WB	0.4	0.4	3.0	7.1	11.5	14.8	17.0	16.2	12.3	9.1	5.1	1.7	(33)
(34)			MCDWB	1.0	1.3	5.1	11.0	16.0	19.6	21.9	19.9	15.0	10.6	6.0	2.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.8	9.0	11.3	12.6	12.7	13.3	12.1	11.6	11.1	8.2	5.9	6.5	(35)
(36)			MCDBR	6.7	9.7	14.1	18.4	18.3	18.6	17.4	15.5	13.5	9.5	7.7	7.3	(36)
(37)			MCWBR	5.9	7.5	9.4	11.2	9.7	9.5	8.5	8.1	8.2	6.8	6.5	6.5	(37)
(38)		5% WB	MCDWR	6.6	8.0	12.4	17.0	15.9	16.0	14.6	13.3	11.6	8.2	7.2	7.4	(38)
(39)			MCWBR	5.9	6.4	8.4	10.5	8.8	8.8	7.5	7.7	7.5	6.3	6.3	6.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.238	0.284	0.312	0.352	0.355	0.347	0.368	0.358	0.326	0.311	0.259	N/A	(40)	
(41)		taud	2.178	2.275	2.302	2.345	2.425	2.471	2.429	2.450	2.528	2.479	2.269	N/A	(41)	
(42)		Ebn at Noon	528	712	813	844	870	883	852	831	803	685	497	N/A	(42)	
(43)		Edh at Noon	37	64	87	102	101	99	101	91	71	54	34	N/A	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.28	0.94	2.44	3.90	4.71	5.32	4.76	3.73	2.46	1.10	0.35	0.17	(44)	
(45)		RadStd	0.04	0.18	0.26	0.46	0.46	0.36	0.66	0.38	0.30	0.21	0.10	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page